### Focus on Evidence

A study by Borstad and Ludewig,[19] which looked at the effect of pectoralis minor resting length on scapular kinematics in subjects without shoulder pain, documented that those individuals with a short pectoralis minor ($n = 25$) had greater scapular internal rotation (protraction) and less posterior tipping during arm elevation in flexion, abduction, and scaption than those with a longer pectoralis minor ($n = 25$), thus providing evidence for altered pectoralis minor muscle length and altered scapular movement. In a related study by the same author,[20] a correlation between the postural impairments of increased thoracic kyphosis, scapular internal rotation and forward tipping, and decreased pectoralis minor length was found to be significant, thus further supporting the relationship between muscle length and posture.

#### Suprahumeral (Subacromial) Space

The coracoacromial arch, composed of the acromion and coracoacromial ligament, overlies the subacromial/subdeltoid bursa, the supraspinatus tendon, and a portion of the muscle (Fig. 17.7).[98] These structures allow for and participate in normal shoulder function. Compromise of this space from faulty muscle function, faulty postural relationships, faulty joint mechanics, injury to the soft tissue in this region, or structural anomalies of the acromion lead to impingement syndromes.[19,27,30,99,103,113] After a rotator cuff tear, the bursa may communicate with the glenohumeral joint cavity.[44]

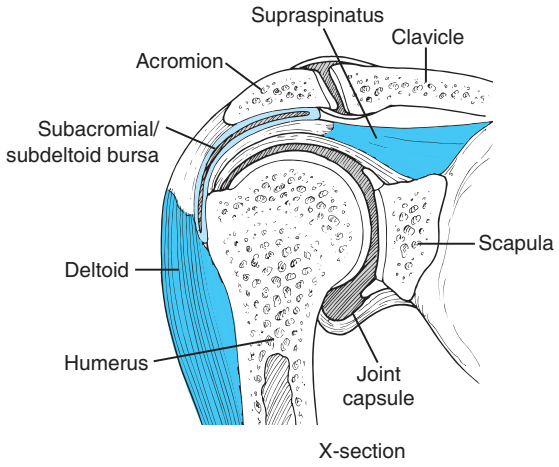



FIGURE 17.7 The supraspinatus and subacromial/subdeltoid bursa lie in the suprahumeral space.

## Shoulder Girdle Function

### Scapulohumeral Rhythm

Motion of the scapula, synchronous with motions of the humerus, allows for 150° to 180° of shoulder ROM into flexion or abduction with elevation. The ratio has considerable variation among individuals but is commonly accepted to be 2:1 (2° of glenohumeral motion to 1° of scapular rotation) overall motion. During the setting phase (0° to 30° abduction, 0° to 60° flexion), motion is primarily at the glenohumeral joint, whereas the scapula seeks a stable position. During the mid-range of humeral motion, the scapula has greater motion, approaching a 1:1 ratio with the humerus; later in the range, the glenohumeral joint again dominates the motion.[39,98,170]

- During humeral elevation, the synchronous motion of the scapula allows the muscles moving the humerus to maintain an effective length–tension relationship throughout the activity and helps maintain congruency between the humeral head and fossa while decreasing shear forces.[39,98,170]
- Muscles causing the upward rotation of the scapula are the upper and lower trapezius and the serratus anterior. Weakness or complete paralysis of these muscles results in the scapula being rotated downward by the contracting deltoid and supraspinatus as abduction or flexion is attempted. These two muscles then reach active insufficiency, and functional elevation of the arm cannot be reached, even though there may be normal passive ROM and normal strength in the shoulder abductor and flexor muscles.[170]
- During elevation of the humerus, the pectoralis minor is lengthened as the scapula upwardly rotates, retracts, and tips posteriorly. Restricted scapular movement during humeral elevation from a shortened pectoralis minor results in patterns similar to those seen in patients with impingement symptoms and could be a risk factor for development of the syndrome.[19]

### Clavicular Elevation and Rotation With Humeral Motion

It is commonly accepted that initially the first 30° of upward rotation of the scapula occurs with elevation of the clavicle at the SC joint. Then, as the coracoclavicular ligament becomes taut, the clavicle rotates 38° to 55° about its longitudinal axis, which elevates its acromial end (because it is crank-shaped). This motion allows the scapula to rotate an additional 30° at the AC joint.[98] Loss of any of these functional components decreases the amount of scapular rotation and thus the ROM of the upper extremity.

### Focus on Evidence

A recent three-dimensional study of clavicular motion during humeral flexion, scaption, and abduction to 115° using surface electromagnetic sensors on 30 asymptomatic subjects and 9 individuals with shoulder pathology documented 11° to 15° of clavicular elevation, 15° to 29° of retraction, and 15° to 31° of posterior long axis rotation, showing similar patterns but different ranges from previously reported studies.[104] Ranges of clavicular motion

above 115° were not reliable owing to movement of the clavicle under the skin.[104]

### External Rotation of the Humerus With Full Elevation

During abduction of the arm in the frontal plane, for the greater tubercle of the humerus to clear the coracoacromial arch, the humerus must externally rotate as it is elevated above the horizontal. Weak or inadequate external rotation results in impingement of the soft tissues in the suprahumeral space, causing pain, inflammation, and eventually loss of function.

### Internal Rotation of the Humerus With Full Elevation

Medial rotation begins at around 50° of passive shoulder flexion when all structures are intact.[136] With full range of shoulder flexion and elevation, the humerus medially rotates 90°, and the medial epicondyle faces anteriorly.[17,18,136]

- As the arm elevates above the horizontal position in the sagittal plane, the anterior capsule and ligaments become taut, causing the humerus to rotate medially. The bony configuration of the posterior aspect of the glenoid fossa contributes to the inward rotation motion of the humerus as the shoulder flexes.[170] Most of the shoulder flexor muscles are also medial rotators of the humerus.[170]
- The infraspinatus and teres minor stabilize the humeral head against the inward rotating forces, helping to maintain alignment and stability of the head in the fossa. Weakness in these muscles may contribute to excessive anterior translation and instability.[31]

### Elevation of the Humerus Through the Plane of the Scapula—Scaption

The plane of the scapula is described as 30° anterior to the frontal plane. Motion of the humerus in this plane is popularly called *scaption*[181,198] or scapular plane abduction.[42] In this range there is less tension on the capsule, and greater elevation is possible than with pure frontal or sagittal plane elevation. Neither internal nor external rotation of the humerus is necessary to prevent greater tubercle impingement during elevation in scaption.[42,181] Many functional activities occur with the shoulder oriented in this plane.

### Deltoid–Short Rotator Cuff and Supraspinatus Mechanisms

- Most of the force of the deltoid muscle causes upward translation of the humerus; if unopposed, it leads to impingement of the soft tissues in the suprahumeral space between the humeral head and the coracoacromial arch.
- The combined effect of the short rotator muscles (infraspinatus, teres minor, subscapularis) causes stabilizing compression and downward translation of the humerus in the glenoid.
- The combined actions of the deltoid and short rotators result in a balance of forces that abduct the humerus and control the humeral head.
- The supraspinatus muscle has a significant stabilizing, compressive, and slight upward translation effect on the humerus; these effects, combined with the effect of gravity, lead to abduction of the arm.
- Interruption of function leading to fatigue or poor coordination of any of these muscles can cause microtrauma and eventual dysfunction in the shoulder region.

## Referred Pain and Nerve Injury

For a detailed description of referred pain patterns, peripheral nerve injuries in the shoulder, thoracic outlet syndrome, and complex regional pain syndromes (including reflex sympathetic dystrophy) and their management, see Chapter 13.

### Common Sources of Referred Pain in the Shoulder Region

#### Cervical Spine

- Vertebral joints between C3 and C4 or between C4 and C5
- Nerve roots C4 or C5

#### Referred Pain from Related Tissues

- Dermatome C4 is over the trapezius to the tip of the shoulder.
- Dermatome C5 is over the deltoid region and lateral arm.
- Diaphragm: pain perceived in the upper trapezius region.
- Heart: pain perceived in the axilla and left pectoral region.
- Gallbladder irritation: pain perceived at the tip of shoulder and posterior scapular region.

### Nerve Disorders in the Shoulder Girdle Region

***Brachial plexus in the thoracic outlet.*** Common sites for compression are the scalene triangle, costoclavicular space and under the coracoid process, and pectoralis minor muscle.[102]

***Suprascapular nerve in the suprascapular notch.*** This injury occurs from direct compression or from nerve stretch, such as when carrying a heavy book bag over the shoulder.

***Radial nerve in the axilla.*** Compression occurs from continual pressure, such as when leaning on axillary crutches.

## MANAGEMENT OF SHOULDER DISORDERS AND SURGERIES

To make sound clinical decisions when managing patients with shoulder disorders, it is necessary to understand the various pathologies, surgical procedures, and associated precautions and to identify presenting impairments, functional limitations, and possible disabilities. In this section, common pathologies and surgeries are presented and are related to corresponding preferred practice patterns (groupings of impairments) described in the *Guide to Physical Therapist Practice*[2] (Table 17.2). Conservative and postoperative management of these conditions are described in this section.

## JOINT HYPOMOBILITY: NONOPERATIVE MANAGEMENT

### Glenohumeral Joint

Restricted mobility of the glenohumeral joint may occur as a result of pathology such as rheumatoid arthritis or osteoarthritis, from prolonged immobilization, or from unknown causes (idiopathic frozen shoulder). Associated impairments in mobility and muscle performance also occur in the muscles and other connective tissues in the area.

#### Related Pathologies and Etiology of Symptoms

***Rheumatoid arthritis and osteoarthritis.*** These disorders follow the clinical picture described in Chapter 11.

TABLE 17.2 Shoulder Pathologies/Surgical Procedures and Preferred Practice Patterns

| PATHOLOGY/SURGICAL PROCEDURE | PREFERRED PRACTICE PATTERNS AND ASSOCIATED IMPAIRMENTS[2] |
|---|---|
| • Abnormal posture (protracted scapula, kyphosis, forward head) | • Pattern 4B—Impaired posture |
| • Arthritis (osteoarthritis, rheumatoid arthritis, traumatic arthritis)<br>• Synovitis<br>• Postimmobilization arthritis (stiff shoulder)<br>• Idiopathic frozen shoulder<br>• Adhesive capsulitis<br>• Joint instability, subluxation, dislocation (nontraumatic/recurrent)<br>• Rotator cuff syndrome and allied disorders<br>• Labral lesion | • Pattern 4D—Impaired joint mobility, motor function, muscle performance, and range of motion (ROM) associated with connective tissue dysfunction |
| • Arthritis—acute stage<br>• Acute impingement syndrome (tendinitis, bursitis)<br>• Acute capsulitis<br>• Acute rotator cuff tear<br>• Traumatic shoulder dislocation | • Pattern 4E—Impaired joint mobility, motor function, muscle performance, and ROM associated with localized inflammation |
| • Fractures (proximal humerus, clavicle, scapula) | • Pattern 4G—Impaired joint mobility, muscle performance, and ROM associated with fracture |
| • Total shoulder replacement<br>• Hemireplacement arthroplasty<br>• Resurfacing and interposition arthroplasties<br>• Resection arthroplasty (distal clavicle) | • Pattern 4H—Impaired joint mobility, motor function, muscle performance, and ROM associated with joint arthroplasty |
| • Subacromial decompression procedures (bursectomy, acromioplasty, distal clavicular resection)<br>• Rotator cuff repair<br>• Capsulorrhaphy (capsular shift)<br>• Electrothermally-assisted arthroscopic capsulorrhaphy<br>• Capsulolabral reconstruction<br>• Fracture stabilization with internal fixation<br>• Joint débridement<br>• Synovectomy<br>• Arthrodesis | • Pattern 4I—Impaired joint mobility, motor function, muscle performance, and ROM associated with bony or soft tissue surgery |
| • Thoracic outlet syndrome (brachial plexus), suprascapular, axillary, median, ulnar, or radial nerve entrapment or injury in the shoulder girdle region. | • Pattern 5F—Impaired peripheral nerve integrity and muscle performance associated with peripheral nerve injury |

***Traumatic arthritis.*** This disorder occurs in response to a fall or blow to the shoulder or to microtrauma from faulty mechanics or overuse.

***Postimmobilization arthritis or stiff shoulder.*** This disorder occurs as a result of lack of movement or secondary effects from conditions such as heart disease, stroke, or diabetes mellitus.

***Idiopathic frozen shoulder.*** This disorder, which is also called *adhesive capsulitis* or *periarthritis,* is characterized by the development of dense adhesions, capsular thickening, and capsular restrictions, especially in the dependent folds of the capsule, rather than arthritic changes in the cartilage and bone, as seen with rheumatoid arthritis or osteoarthritis. The onset is insidious and usually occurs between the ages of 40 and 60 years; there is no known cause (primary frozen shoulder), although problems already mentioned in which there is a period of pain and/or restricted motion, such as with rheumatoid arthritis, osteoarthritis, trauma, or immobilization, may lead to a frozen shoulder (secondary frozen shoulder). With primary frozen shoulder, the pathogenesis may be a provoking chronic inflammation in musculotendinous or synovial tissue such as the rotator cuff, biceps tendon, or joint capsule.[41,66,122,125] Consistent with this is a faulty posture and muscle imbalance predisposing the suprahumeral space to impingement and overuse syndromes.[1]

### Clinical Signs and Symptoms

***Glenohumeral joint arthritis.*** The following characteristics are associated with glenohumeral (GH) joint pathologies that lead to hypomobility.

- *Acute phase.* Pain and muscle guarding limit motion, usually external rotation and abduction. Pain is frequently experienced radiating below the elbow and may disturb sleep. Joint swelling is not detected owing to the depth of the capsule, although tenderness can be elicited by palpating in the fornix immediately below the edge of the acromion process between the attachments of the anterior and middle deltoid.
- *Subacute phase*. Capsular tightness begins to develop. Limited motion is detected, consistent with a capsular pattern (external rotation and abduction are most limited, and internal rotation and flexion are least limited). Often, the patient feels pain as the end of the limited range is reached. Joint-play testing reveals limited joint play. If the patient can be treated as the acute condition begins to subside by gradually increasing shoulder motion and activity, the complication of joint and soft tissue contractures can usually be minimized.[117,122]
- *Chronic phase.* Progressive restriction of the GH joint capsule magnifies the signs of limited motion in a capsular pattern and decreased joint play. There is significant loss of function with an inability to reach overhead, outward, or behind the back. Aching is usually localized to the deltoid region.

***Idiopathic frozen shoulder.*** This clinical entity follows a classic pattern*.

- *"Freezing."* Characterized by intense pain even at rest and limitation of motion by 2 to 3 weeks after onset. These acute symptoms may last 10 to 36 weeks.
- *"Frozen."* Characterized by pain only with movement, significant adhesions, and limited GH motions, with substitute motions in the scapula. Atrophy of the deltoid, rotator cuff, biceps, and triceps brachii muscles occurs. This stage lasts 4 to 12 months.
- *"Thawing."* Characterized by no pain and no synovitis but significant capsular restrictions from adhesions. This stage lasts 2 to 24 months or longer. Some patients never regain normal ROM.

Some references indicate that spontaneous recovery occurs, on average, 2 years from onset,[66] although others have reported long-term limitations without spontaneous recovery.[156] Inappropriately aggressive therapy at the wrong time may prolong the symptoms.[15] Management guidelines are the same as for acute (maximum protection during the freezing stage), subacute (controlled motion during the frozen stage), and chronic (return to function during the thawing state) joint pathology described in this section.

### Common Impairments

- Night pain and disturbed sleep during acute flares
- Pain on motion and often at rest during acute flares
- Mobility: decreased joint play and ROM, usually limiting external rotation and abduction with some limitation of internal rotation and elevation in flexion
- Posture: possible faulty postural compensations with protracted and anteriorly tipped scapula, rounded shoulders, and elevated and protected shoulder
- Decreased arm swing during gait
- Muscle performance: general muscle weakness and poor endurance in the glenohumeral muscles with overuse of the scapular muscles leading to pain in the trapezius and posterior cervical muscles
- Guarded shoulder motions with substitute scapular motions

### Common Functional Limitations/Disabilities

- Inability to reach overhead, behind head, out to the side, and behind back; thus, having difficulty dressing (such as putting on a jacket or coat or women fastening undergarments behind their back), reaching hand into back pocket of pants (to retrieve wallet), reaching out a car window (to use an ATM machine), self-grooming (such as combing hair, brushing teeth, washing face), and bringing eating utensils to the mouth
- Difficulty lifting weighted objects, such as dishes into a cupboard
- Limited ability to sustain repetitive activities

*See references 41, 66, 122, 125, 153, 189

### Glenohumeral Joint Hypomobility: Management—Protection Phase

See general guidelines for management when symptoms are acute in Chapter 10 and Box 10.1.

#### Control Pain, Edema, and Muscle Guarding

- The joint may be immobilized in a sling to provide rest and minimize pain.
- Intermittent periods of passive or assisted motion within the pain free/protected ROM and gentle joint oscillation techniques are initiated as soon as the patient tolerates movement in order to minimize adhesion formation.

#### Maintain Soft Tissue and Joint Integrity and Mobility

- *Passive range of motion* (PROM) in all ranges of pain-free motion (see Chapter 3). As pain decreases, the patient is progressed to active ROM with or without assistance using activities such as rolling a small ball or sliding a rag on a smooth table top in flexion, abduction, and circular motions. Be sure the patient is taught proper mechanics and avoids faulty patterns, such as scapular elevation or a slumped posture.
- *Passive joint distraction and glides, grade I and II* with the joint placed in a pain-free position (see Chapter 5).
- *Pendulum (Codman's) exercises* are techniques that use the effects of gravity to distract the humerus from the glenoid fossa.[30,34] They help relieve pain through gentle traction and oscillating movements (grade II) and provide early motion of joint structures and synovial fluid. No weight is used during this phase of treatment (see Fig. 17.22).

PRECAUTION: If there is increased pain or irritability in the joint after use of these techniques, either the dosage was too strong or the techniques should not be used at this time.

CONTRAINDICATION: Stretching (grade III) techniques. If there are mechanical restrictions causing limited motion, appropriate stretching can be initiated only *after* the inflammation subsides.

- *Gentle muscle setting* to all muscle groups of the shoulder, including scapular and elbow muscles because of their close association with the shoulder. Instructions are given to the patient to gently contract a group of muscles while slight resistance is applied—just enough to stimulate a muscle contraction. It should not provoke pain. The emphasis is on rhythmic contracting and relaxing of the muscles to help stimulate blood flow and prevent circulatory stasis.

#### Maintain Integrity and Function of Associated Areas

- Reflex sympathetic dystrophy (complex regional pain syndrome type I) is a potential complication after shoulder injury or immobility. Therefore special attention is given to the hand with additional exercises, such as having the patient repetitively squeeze a ball or other soft object.
- The patient is advised of the importance of keeping the joints distal to the injured site as active and mobile as possible. The patient or family member is taught to perform ROM exercises of the elbow, forearm, wrist, and fingers several times each day while the shoulder is immobilized. If tolerated, active or gentle resistive ROM is preferred to passive ROM for a greater effect on circulation and muscle integrity.
- If edema is noted in the hand, instruct the patient to elevate the hand, whenever possible, above the level of the heart.

NOTE: Conditions in which there is potentially a prolonged acute/inflammatory stage, such as with rheumatoid arthritis and during the freezing stage of idiopathic frozen shoulder, it is critical to teach the patient active-assistive exercises to maintain muscle and joint integrity and as much mobility as possible without exacerbating the symptoms.

### GH Joint Hypomobility: Management—Controlled Motion Phase

When symptoms are subacute, follow the guidelines as described in Chapter 10, Box 10.2, emphasizing joint mobility, neuromuscular control, and instructions to the patient for self-care.

NOTE: For normal shoulder joint mechanics, there must be good scapular posture and control, and the humerus must be able to externally rotate. To avoid suprahumeral impingement, passive stretching above 90° should be avoided until there is adequate glenohumeral external rotation. With a traumatic injury that involves the AC or SC joints, these joints tend to become hypermobile with improper stretching. Care should be taken to provide stabilization to the scapula and clavicle so as not to stretch these joints when mobilizing the glenohumeral joint.

#### Control Pain, Edema, and Joint Effusion

- *Functional activities.* It is important to carefully monitor increasing activities. If the joint was splinted, the amount of time the shoulder is free to move each day is progressively increased.
- *Range of motion.* ROM is progressed up to the point of pain, including all shoulder and scapular motions. The patient is instructed in the use of self-assistive ROM techniques, such as the wand exercises or hand slides on a table.

PRECAUTION: With increased pain or decreased motion, the activity may be too intense or the patient may be using faulty mechanics. Reassess the technique and modify it if faulty joint mechanics exist.

#### Progressively Increase Joint and Soft Tissue Mobility

- *Passive joint mobilization techniques.* Stretch grades (grade III sustained or grade III and IV oscillation) using techniques that focus on the restricting capsular tissue at the end of the available ROM are used to increase joint capsule mobility[89,127,187] (see Box 17.1 and Figs. 5.15 through 5.20 in Chapter 5). End-of-range techniques

include rotating the humerus and then applying either a grade III distraction or a grade III glide to stretch the restrictive capsular tissue or adhesions (see Figs. 5.17, 5.21, and 17.20).

Use a grade I distraction with all gliding techniques. If the joint is highly irritable and gliding in the direction of restriction is not tolerated, glide in the opposite direction. As pain and irritability decrease, begin to glide in the direction of restriction.[89]

### Focus on Evidence

Evidence supporting joint mobilization techniques is limited. A multiple-subject case study, using seven subjects with adhesive capsulitis of the glenohumeral joint (mean disease duration 8.4 months, range 3–12 months) treated with end-range mobilization techniques twice a week for 3 months, showed increased active and passive range and increased capacity of the joint capsule at the end of treatment and at the 9-month follow-up. No control groups were used; therefore the natural course of the disease could not be excluded as the explanation for improvement.[187]

A follow-up study by the same author randomly assigned 100 subjects with stage II adhesive capsulitis to a group receiving high-grade mobilization techniques (end-range stretching using Maitland grade III or IV) or a group receiving low-grade mobilization techniques (Maitland grade I or II in nonstressful positions). After 3 months of treatment, both groups exhibited clinically significant improvement, with the group receiving the high-grade mobilization techniques showing greater improvement than the low-grade mobilization group. Because there was no control group, natural progression could not be ruled out.[188]

- Pendulum exercises can also be used for joint stretching by adding a cuff weight to the wrist or a weight to the hand to cause a grade III joint distraction force (see Fig. 17.22). To direct the stretch force to the glenohumeral joint, stabilize the scapula against the thorax manually or with a belt.

PRECAUTION: Carefully monitor the joint reaction to the mobilization stretches; if irritability increases, vigorous stretching should not be undertaken until the chronic stage of healing.

- *Self-mobilization techniques.* The following self-mobilization techniques may be used for a home program.
  - CAUDAL GLIDE. *Patient position and procedure:* Sitting on a firm surface and grasping the fingers under the edge. The patient then leans the trunk away from the stabilized arm (Fig. 17.8).
  - ANTERIOR GLIDE. *Patient position and procedure:* Sitting with both arms behind the body or lying supine supported on a solid surface. The patient then leans the body weight between the arms (Fig. 17.9).
  - POSTERIOR GLIDE. *Patient position and procedure:* Prone, propped up on both elbows. The body weight shifts downward between the arms (Fig. 17.10).

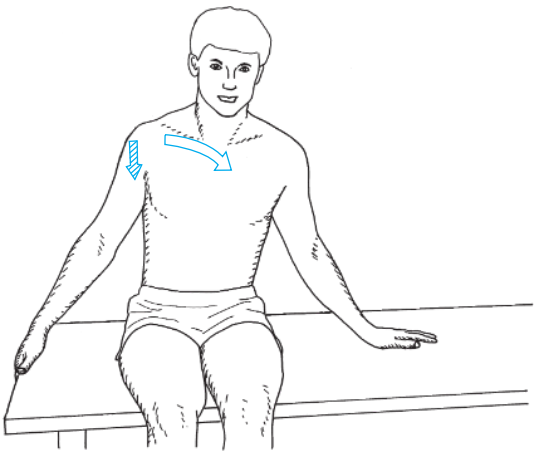


FIGURE 17.8 Self-mobilization. Caudal glide of the humerus occurs as the person leans away from the fixed arm.

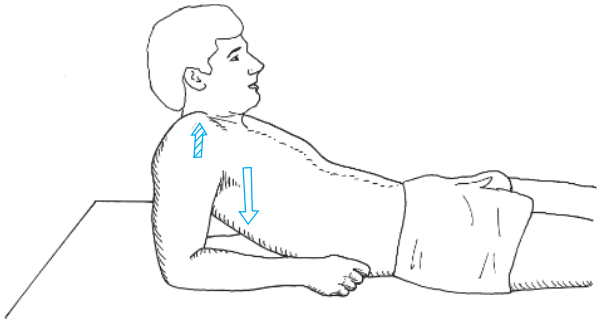


FIGURE 17.9 Self-mobilization. Anterior glide of the humerus occurs as the person leans between the fixed arms.

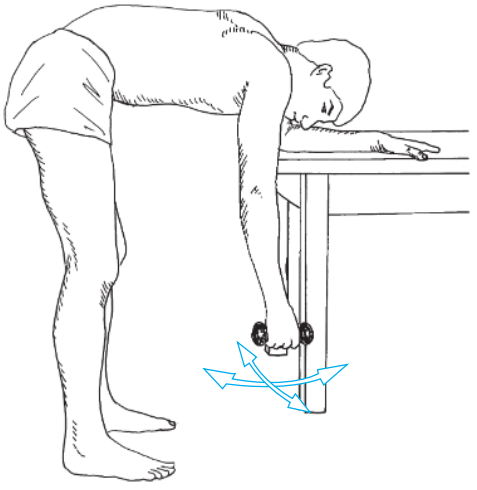


FIGURE 17.10 Self-mobilization. Posterior glide of the humerus occurs as the person shifts his weight downward between the fixed arms.

- *Manual stretching.* Manual stretching techniques are used to increase mobility in shortened muscles and related connective tissue.
- *Self-stretching exercises.* As the joint reaction becomes predictable and the patient begins to tolerate stretching, self-stretching techniques are taught (see Figs. 17.24 through 17.29 in the exercise section).

### Inhibit Muscle Spasm and Correct Faulty Mechanics

Muscle spasm may lead to a faulty deltoid–rotator cuff mechanism and scapulohumeral rhythm when the patient attempts abduction (Fig. 17.11). The head of the humerus may be held in a cranial position in the joint, making it difficult and/or painful to abduct the shoulder because the greater tuberosity impinges on the coracoacromial arch. In this case, repositioning the head of the humerus with a caudal glide is necessary before proceeding with any other form of shoulder exercise. The patient also needs to learn to avoid "hiking the shoulder" when abducting or flexing the arm. The following techniques may address these problems and faulty mechanics. See also mobilization with movement techniques in the next section.

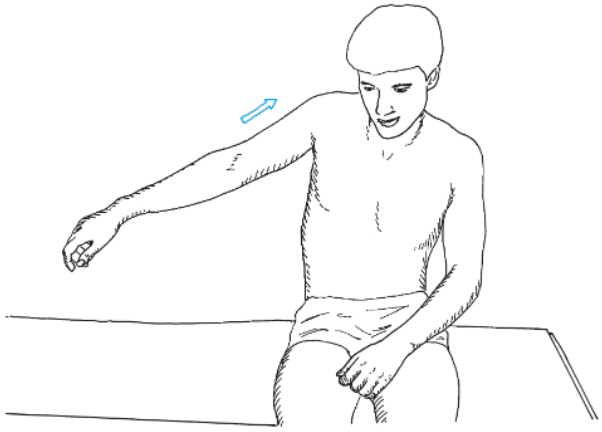


FIGURE 17.11 Poor mechanics with patient hiking the shoulder while trying to abduct the shoulder, thus elevating rather than depressing the humeral head.

- Gentle joint oscillation techniques to help decrease the muscle spasm (grade I or II).
- Sustained caudal glide joint techniques to reposition the humeral head in the glenoid fossa.
- Protected weight bearing, such as leaning hands against a wall or on a table, to stimulate co-contraction of the rotator cuff and scapular stabilizing muscles. If tolerated, gentle rocking forward/backward and side to side requires the muscles to begin controlling motion. Because weight bearing causes joint compression, the benefits of intermittent compression stimulates synovial fluid motion. Techniques are progressed within the tolerance of the joint.
- External rotation exercises to help to depress the humeral head (see Fig. 17.50).

### Improve Joint Tracking

Mobilization with movement (MWM) techniques may assist with retraining muscle function for proper tracking of the humeral head.[116]

- *Shoulder MWM for painful restriction of shoulder external rotation* (Fig. 17.12).
  - *Patient position:* Supine lying with folded towel under the scapula; the elbow is near the side and flexed to 90°. A cane is held in both hands.
  - *Therapist position and procedure:* Stand on the opposite side of the bed facing the patient and reach across the patient's torso to cup the anteromedial aspect of the head of the humerus with reinforced hands. Apply a pain-free graded posterolateral glide of the humeral head on the glenoid. Instruct the patient to use the cane to push the affected arm into the previously restricted range of external rotation. Sustain the movement for 10 seconds and repeat in sets of 5 to 10. It is important to maintain the elbow near the side of the trunk and ensure that no pain is experienced during the procedure. Adjust the grade and direction of the glide as needed to achieve pain-free function.

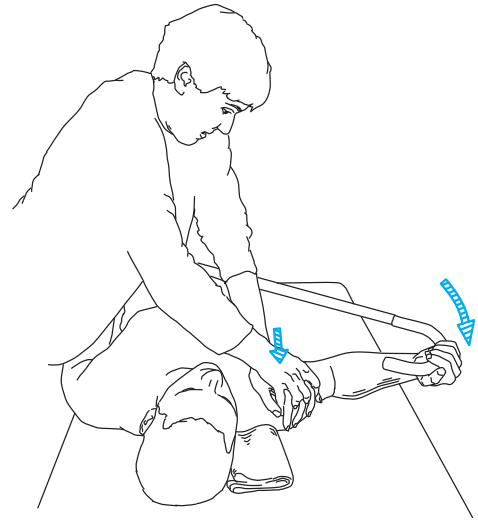


FIGURE 17.12 Mobilization with movement (MWM) to improve external rotation. A posterolateral glide is applied to the humeral head while the patient pushes the arm into the end-range of external rotation with a cane.

- *Shoulder MWM for painful restriction of internal rotation* and inability to reach the hand behind the back (Fig. 17.13).
  - *Patient position:* Standing with a towel draped over the unaffected upper trapezius and affected hand at current range of maximum pain-free position behind back. The patient's hand on the affected side grasps the towel behind the back.
  - *Therapist position and procedure:* Stand facing the patient's affected side. Place the hand closest to the patient's back high up in the axilla with the palm facing outward to stabilize the scapula with an upward

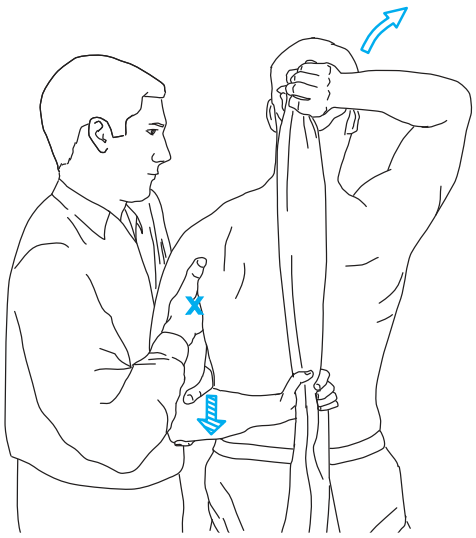


FIGURE 17.13 MWM to improve internal rotation. An inferior glide is applied to the humerus while the patient pulls the hand up the back with a towel.

and inward pressure. With the hand closest to the patient's abdomen hook the thumb in the cubital fossa and grasp the lower humerus to provide an inferior glide. Your abdomen is in contact with the patient's elbow to provide an adduction force to the arm. Have the patient pull on the towel with the unaffected hand to draw the affected hand up the back while the mobilization force is being applied in an inferior direction. Ensure that no pain is experienced during the procedure. Adjust the grade and direction of glide as needed to achieve pain-free function. Maximal glide should be applied to achieve end-range loading.

- *Shoulder MWM for painful arc or impingement signs.* If impingement signs are present in addition to the capsular restrictions, the MWM active elevation technique may be appropriate; see description in the impingement section and Figure 17.17

### Improve Muscle Performance

- Any faulty postures or shoulder girdle mechanics the patient displays when moving the upper extremity in various functional patterns should be identified and exercises to correct the muscle imbalances initiated. Stabilization, flexibility, and strengthening exercises for the shoulder-girdle musculature are described in the last section of this chapter. Emphasis is on developing control in the weak musculature. As the patient learns to activate the weak muscles, progress to strengthening functional patterns of motion.
- Exercises to manage faulty spinal posture are described in Chapter 16.
- Once proper mechanics are restored, the patient should perform active ROM of all shoulder motions daily and return to functional activities to the extent tolerated.

#### GH Joint Hypomobility: Management—Return to Function Phase

For joint impairments in the chronic stage, follow the guidelines described in Chapter 10, Box 10.4.

### Progressively Increase Flexibility and Strength

- Stretching and strengthening exercises are progressed as the joint tissue tolerates. The patient should be actively involved in self-stretching and strengthening by this time, so emphasis during treatment is on correct mechanics, safe progressions, and exercise strategies for return to function.
- If capsular tissue is still restricting ROM, vigorous manual stretching and joint mobilization techniques are applied.

### Prepare for Functional Demands

If the patient is involved in repetitive heavy lifting, pushing, pulling, carrying, or reaching, when joint range and strength allow, exercises are progressed to replicate these demands. See the last section of this chapter for suggestions.

#### GH Joint Management—Postmanipulation Under Anesthesia

Occasionally, no progress is made, and the physician chooses to perform manipulation under anesthesia. Following this procedure, there is an inflammatory reaction and the joint is treated as an acute lesion. If possible, joint-play and passive ROM techniques are initiated while the patient is still in the recovery room. Surgical intervention with incision of the dependent capsular fold may be used if the adhesions are not broken with the manipulation. Postoperative treatment is the same with the following considerations.[125]

- The arm is kept elevated overhead in abduction and external rotation during the inflammatory reaction stage; treatment principles progress as with any joint lesion.
- Therapeutic exercises are initiated the same day while the patient is still in the recovery room, with emphasis on internal and external rotation in the 90° (or higher) abducted position.
- Joint mobilization procedures are used, particularly a caudal glide, to prevent readherence of the inferior capsular fold.
- When sleeping, the patient may be required to position the arm abducted for up to 3 weeks after manipulation.

## Acromioclavicular and Sternoclavicular Joints

### Related Pathologies and Etiology of Symptoms

***Overuse Syndromes.*** Overuse syndromes of the AC joint are frequently arthritic or post-traumatic conditions. The causes may be from repeated stressful movement of the joint with the arm at waist level, such as with grinding, packing assembly, and construction work,[67] or repeated diagonal extension, adduction, and internal rotation motions, as when spiking a volleyball or serving in tennis.

***Subluxations or Dislocations.*** Subluxations or dislocations of either joint are usually caused by falling against the shoulder or against an outstretched arm. In the AC joint, the distal end of the clavicle displaces posteriorly and superiorly on the acromion; the ligaments supporting the AC joint may rupture.[124] Clavicular fractures may result from the fall.[124] After trauma and associated overstretching of the capsules and ligaments of either joint, hypermobility is usually permanent because there is no muscle support to restrict movement.

***Hypomobility.*** Decreased clavicular mobility may occur with sustained faulty postures involving clavicular and scapular depression or retraction. Restricted mobility may contribute to a thoracic outlet syndrome (TOS) with a compromise of space for the neuromuscular bundle as it courses between the clavicle and first rib (described in Chapter 13).

### Common Impairments

- Pain localized to the involved joint or ligament
- Painful arc with shoulder elevation
- Pain with shoulder horizontal adduction or abduction
- Hypermobility in the joints if trauma or overuse is involved
- Hypomobility in the joints if sustained posture or immobility is involved
- Neurological or vascular symptoms if TOS is present

### Common Functional Limitations/Disabilities

- Limited ability to sustain repeated loaded movements related to forward/backward motions of the arm, such as with grinding, packing, assembly, and construction work.[67]
- Inability to reach overhead without pain.
- Inability to serve effectively at tennis or spike a volleyball.
- See also limitations/disabilities from TOS if present (see Chapter 13).

### Nonoperative Management of AC or SC Joint Strain or Hypermobility

- Rest the joint by putting the arm in a sling to support the weight of the arm.
- Cross-fiber massage to the capsule or ligaments.
- ROM to the shoulder and grade II traction and glides to the glenohumeral joint to prevent glenohumeral restriction.
- Instructions in self-application of cross-fiber massage if joint symptoms occur after excessive activity.

### Nonoperative Management of AC or SC Joint Hypomobility

Joint mobilization techniques are used to increase joint mobility (see Figs. 5.22 through 5.24).

## Glenohumeral Joint Surgery and Postoperative Management

Severe deterioration of one or both surfaces of the GH joint, causing significant pain and loss of upper extremity function, or an acute or nonunion fracture of the proximal humerus often must be addressed with surgical intervention. Underlying pathologies, causing advanced joint destruction, include late-stage osteoarthritis (OA), rheumatoid arthritis (RA), traumatic arthritis, cuff tear arthropathy, and osteonecrosis (avascular necrosis) of the head of the humerus as the result of a fracture of the anatomical neck of the humerus or long-term use of steroids for systemic disease.

The most common surgical procedure used to treat advanced shoulder joint pathology is *glenohumeral arthroplasty,* often simply referred to as *shoulder arthroplasty.*[36] In rare situations, *arthrodesis* (surgical ankylosis) of the GH joint may have to be selected as an alternative to arthroplasty or as a salvage procedure.[109]

The goals of these surgical procedures and postoperative rehabilitation are to (1) relieve pain, (2) improve shoulder mobility or stability, and (3) restore or improve strength and functional use of the upper extremity. The extent to which these goals are achieved is predicated on the patient's participation in postoperative rehabilitation, the distinguishing features and severity of the underlying pathology, the prosthetic design and surgical techniques, the integrity of the rotator cuff mechanism and other soft tissues, and the age, overall health, and anticipated activity level of the patient.[36,109,161,168]

### Glenohumeral Arthroplasty

Arthroplasty of the GH joint falls into several categories, the most common of which are *total shoulder replacement arthroplasty*,[109,119,161,168] in which the glenoid and humeral surfaces are replaced (Fig. 17.14), and *hemireplacement arthroplasty (hemiarthroplasty)*, in which one surface, the humeral head, is replaced.[6,57,109,121,161] Other categories of shoulder arthroplasty include interpositional and resurfacing arthroplasties, which involve less extensive removal of bone.[109,161,168,182]

#### Indications for Surgery

The following impairments associated with these pathologies are widely accepted indications for GH arthroplasty.*

- The primary indication is persistent and incapacitating pain (at rest or with activity) secondary to GH joint destruction.

*See references 6, 36, 47, 108, 109, 119, 121, 161, 168, 173, 174, 195.



FIGURE 17.14 Postoperative anterior-posterior view of the shoulder showing a Neer II type of cemented humeral prosthesis and a nonmetal backed polyethylene glenoid. *(From Tovin, BJ, Greenfield, BH: Evaluation and Treatment of the Shoulder—An Integration of the Guide to Physical Therapist Practice. FA Davis, Philadelphia, 2001, p 266, with permission.)*

- Secondary indications include loss of shoulder mobility or stability and upper extremity strength leading to inability to perform functional tasks with the involved upper extremity.

### Procedures

#### Background

***Implant design, materials, and fixation.*** Since the pioneering work of Neer during the 1960s and 1970s[119,121] and many other investigators,[28] prosthetic designs and surgical techniques for replacing the shoulder joint have continued to evolve. The designs of current-day total shoulder replacement (TSR), composed of a high-density polyethylene glenoid component (usually all plastic) and a modular humeral component made of an inert metal, closely approximate the biomechanical characteristics of the human shoulder.[206]

Fixation of the prosthetic components is achieved with a press fit, bio-ingrowth, or cement. The type of fixation selected by the surgeon depends on the component (glenoid or humeral), the underlying pathology, and the quality of the bone stock. Cement fixation is most often necessary in patients with osteoporosis.†

†See references 36, 108, 109, 173, 206.

**BOX 17.3 Designs of Prosthetic Implants for Total Shoulder Replacement**

**Unconstrained**
- Anatomical design with a small, shallow glenoid component combined with a stemmed humeral component
- The most frequently used prosthetic design
- Provides the greatest freedom of shoulder motion but no inherent stability
- Indicated when the rotator cuff mechanism is intact or can be repaired to provide dynamic stability to the GH joint

**Semiconstrained**
- A larger glenoid component that is hooded or cup-shaped
- Some degree of joint stability inherent in the design
- Indicated when erosion of the glenoid fossa can be compensated for by reaming the fossa and rotator cuff function; although deficient preoperatively, can be improved by repair

**Reversed ball and socket**
- Small humeral socket that slides on a larger ball-shaped glenoid component
- Couples some degree of stability with mobility for rotator cuff-deficient shoulders that cannot be repaired
- Provides an alternative to standard semiconstrained and total shoulder replacement (TSR) and hemiarthroplasty

**Constrained**
- Fixed fulcrum, ball-in-socket designs with congruency of the glenoid and humeral components
- Greatest amount of inherent joint stability but less mobility than less constrained designs
- Once thought to be an alternative to hemiarthroplasty for the selected patient with a deficient rotator cuff or cuff tear arthropathy or chronic/recurrent GH joint dislocation after a previous TSR
- Rarely used today owing to high rate of loosening or failure of the components

The designs of total shoulder replacements, ranging from *unconstrained* to *semiconstrained* (including *reverse ball-and-socket*) to *constrained,* provide varying amounts of mobility and stability of the GH joint. Box 17.3 summarizes the characteristics of each of these designs.[36,108,109,159,161,168,182]

NOTE: The description of constrained designs is included in Box 17.3 for historical purposes and for a comparison with less constrained designs. Because of the high rate of complications that occurred with constrained designs, these implant systems are rarely, if ever, used today.[36,109]

***Selection of procedure.*** Controversy exists over the specific criteria for selection of TSR versus hemiarthroplasty,

but in general it depends on the etiology and severity of the joint deterioration and the condition of the periarticular soft tissues, particularly the rotator cuff mechanism.[109,169] Several examples that follow underscore the complexity of the clinical decision-making process involved in the choice of operative procedure and prosthetic design.

In patients with late-stage primary OA, the most common pathology associated with shoulder arthroplasty, the GH joint typically exhibits loss or thinning or the articular cartilage of the head of the humerus and the posterior portion of the glenoid fossa. In addition, the rotator cuff is intact in approximately 90% to 95% of these patients, making them good candidates for either TSR or hemiarthroplasty.[36,108,139,159,161,168] However, opinions vary on whether selection of an unconstrained TSR yields results that are better than or equal to hemiarthroplasty for shoulders with these characteristics.[36,109,134,161,171]

Chronic synovitis, asssociated with RA and other types of synovium-based arthritis, tends to erode periarticular soft tissues in addition to the articular surfaces of a joint. As a consequence, a rupture or full-thickness tear of a rotator cuff tendon (typically the supraspinatus) develops in 25% to 40% of these patients and a rupture of the biceps tendon in an even greater percentage.[57,159,168,173,182] If the soft tissues can be repaired and their functions improved, a semiconstrained TSR (standard or possibly a reverse ball-and-socket design) that may include bone grafting at the glenoid to improve prosthetic fixation may be indicated. If an effective cuff repair cannot be achieved or if there is insufficient bone stock owing to osteoporosis for fixation of a glenoid implant, hemiarthroplasty is usually the procedure of choice.[57,109,159,161,169,173,182]

Hemiarthroplasty is often used when the articular surface and underlying bone of the humeral head have deteriorated but the glenoid fossa is reasonably intact, as seen with osteonecrosis of the head of the humerus.[36,109,168,169] A patient with severe, chronic pain and loss of function as the result of a massive, irreparable cuff tear and subsequent development of a *cuff tear arthropathy* is typically a candidate for hemiarthroplasty.[57,159,212] (The term "cuff tear arthropathy," first used by Neer, refers to deterioration and eventual collapse of the head of the humerus, an infrequent but debilitating long-term result of a primary, massive and irreparable tear of the rotator cuff.[109,159,195]) Chronic deficiency of the rotator cuff mechanism leads to superior migration of the head of the humerus in the glenoid fossa. If a glenoid component is inserted under these conditions, the superior migration creates an incongruous articulation that accentuates the risk of loosening and premature wear of the glenoid implant.[35,73,159]

### Operative Procedures

Total shoulder replacement and hemiarthroplasty are open surgical procedures performed with the patient in a semi-reclining position. The operative procedure involves the following components*: anterior approach using a deltopectoral incision that extends from the AC joint to the deltoid insertion for adequate surgical exposure; release (tenotomy) of the subscapularis tendon from its proximal attachment on the lesser tuberosity; anterior capsulotomy; exposure of the humeral head for a humeral osteotomy; and preparation of the humeral canal for insertion of the prosthetic implant. The glenoid fossa is débrided and for a TSR is precisely contoured so the glenoid implant can be placed flush within the fossa.

Reconstruction and balancing of soft tissues is critical for optimal function after TSR and hemiarthroplasty. "Balancing" refers to the intraoperative lengthening or tightening of soft tissues to restore as near-normal resting tension in the tissues as possible, particularly in the rotator cuff, biceps, and deltoid muscle-tendon units.

Concomitant procedures that may be necessary during shoulder arthroplasty include:

- Repair of a deficient rotator cuff if the quality of the cuff tissue is sufficient
- Repair (reattachment) of the subscapularis and lengthening (medial advancement or Z-plasty) if a contracture significantly limits external rotation
- Capsular plication and tightening for chronic subluxation or dislocation (usually posterior) of the GH joint
- Anterior acromioplasty for a history of impingement syndrome
- Bone graft of the glenoid if bone stock is insufficient for fixation of the glenoid implant

After implantation of the prosthetic component(s) and repair of soft tissues but before closure of the skin incision, the shoulder is passively moved through all planes of motion to visually evaluate the stability of the prosthetic joint and the integrity of the repaired soft tissues. This determines the anatomical ROM possible after surgery and how aggressive the postoperative program can be.[36,109]

### Complications

Complications that can occur after a wide variety of orthopedic surgeries are summarized in Chapter 12 (see Box 12.6). Although the incidence of intraoperative and postoperative complications after current-day arthroplasty is low, even a single complication can adversely affect the functional outcome. The incidence of complications tends to be higher in patients with a deficient rotator cuff mechanism, osteoporosis, and a preoperative history of chronic GH joint instability.[72] Aside from medical complications, such as infection or a deep vein thrombosis, complications specific to shoulder arthroplasty are noted in Box 17.4.[35,72]

## Postoperative Management

NOTE: Effective patient education and close communication among the therapist, surgeon, and patient are the basis of an effective and safe rehabilitation program individualized to address the specific surgical procedures used to meet the unique needs of each patient.

### Special Considerations

***Integrity of the rotator cuff.*** Regardless of the underlying cause of late-stage glenohumeral arthritis, the goals,

*See references 6, 36, 57, 109, 161, 168.

**BOX 17.4 Complications Specific to Glenohumeral Arthroplasty**

**Intraoperative**
- Insufficient lengthening of a tight subscapularis muscle-tendon unit
- Intraoperative damage to the axillary or suprascapular nerves, affecting the deltoid and supraspinatus/infraspinatus muscles, respectively
- Fracture of the humerus

**Soft Tissue–Related Postoperative Complications**
- Re-tearing a repaired rotator cuff mechanism
- Postoperative disruption of the repaired subscapularis (detached from the lesser tuberosity for the surgical approach and reattached medially on the cut surface of the neck of the humerus)
- Chronic instability of the GH joint
- Progressive erosion of the articular surface of the glenoid fossa (after hemiarthroplasty)

**Implant-Related Postoperative Complicatons**
- After TSR mechanical (aseptic) loosening or premature wear or fracture of the polyethylene glenoid implant
  - Most often seen in a rotator cuff-deficient shoulder
  - Due to excessive stresses at the bone–prosthesis interface
  - Low incidence with unconstrained designs but higher with early-generation constrained designs
- Loosening of the humeral prosthesis after hemiarthroplasy

components, and rate of progression of a rehabilitation program after TSR or hemiarthroplasty are influenced by the pre- and postoperative integrity of the rotator cuff mechanism. A deficient rotator cuff compromises the dynamic stability and kinematics of the GH joint, as demonstrated by upward riding of the humeral head leading to increased shear forces at the joint during active elevation of the arm.[43] The rehabilitation program for a patient with an intact rotator cuff prior to shoulder arthroplasty can be progressed more rapidly than the program for a patient with coexisting rotator cuff deficiency requiring a concomitant cuff–tendon repair at the time of shoulder arthroplasty. The postoperative program for a patient with an irreparable rotator cuff or a tenuous repair due to poor tendon quality must be progressed even more cautiously.[43,47,92,109,161,195]

If the rotator cuff was intact prior to surgery, the emphasis of postoperative rehabilitation is to restore shoulder mobility and functional use of the arm as soon as possible while protecting soft tissues as they heal. In contrast, with an irreparable rotator cuff, a tenuous repair, or a preoperative history of recurrent GH dislocation, rehabilitation must place greater emphasis on improving or maintaining joint stability for functional use of the arm than on increasing shoulder mobility.[43,47,51,92,109]

***Intraoperative ROM***. In the operating room before closing the incision, the surgeon determines the extent of shoulder ROM available, particularly elevation and external rotation, that does not place undue stress on periarticular soft tissues or compromise GH joint stability. Subsequent goals for safe, stable postoperative ROM are based on these measurements. For a patient with an unconstrained TSR and sufficient postoperative shoulder stability (static and dynamic), the goal at the conclusion of rehabilitation is to achieve active ROM equal to intraoperative ROM—ideally, 140° to 150° of shoulder elevation and 45° to 50° of external rotation.[43,109] For a patient with more constrained TSR, a deficient rotator cuff mechanism, or capsuloligamentous laxity, intraoperative ROM is typically less, and postoperative goals focus more on developing dynamic stability and less on shoulder mobility.

***Posture.*** Many patients undergoing shoulder arthroplasty are elderly. Therefore, the postural changes associated with aging[95] (increased thoracic kyphosis and scapular protraction) cause malalignment (excessive downward rotation) of the glenoid fossa, which predisposes the patient to shoulder impingement and pain particularly during overhead shoulder motions. Accordingly, it is important to emphasize an erect sitting or standing posture during elevation of the arm and to incorporate spinal extension and scapular retraction exercises into the postoperative program.

### Immobilization and Postoperative Positioning

At the close of the surgical procedure the operated arm is placed in some type of shoulder immobilizer, usually a sling or sometimes a splint, to protect reattached and repaired soft tissues and for comfort.[6,36,109,159,168,174] Early postoperative positioning that protects the operated shoulder is detailed in Box 17.5.

Initially, the sling or splint is removed only for exercise and bathing. A patient who did not require repair of

**BOX 17.5 Positioning After Shoulder Arthroplasty: Early Postoperative (Maximum Protection) Phase**

**Supine**
- Arm immobilized in sling, which is worn continuously
  - Elbow flexed to 90°
  - Forearm and hand resting on abdomen
- Arm supported at the elbow on a folded blanket or pillow slightly away from the side and anterior to the midline of the trunk
  - Forward flexion (10° to 20°), slight abduction, and internal rotation of the shoulder
  - Head of bed elevated about 30°

**Sitting**
- Arm supported in sling or resting on a pillow in the patient's lap or on the armrest of a chair

**With Tenuous Rotator Cuff Repair**
- In some cases if a sling does not provide adequate protection of a repaired cuff, an abduction splint must be worn

the rotator cuff is weaned from the sling during the day as quickly as possible to prevent postoperative stiffness. However, a patient who underwent a cuff repair or other soft tissue reconstruction may need to wear a sling or splint while out in crowded areas or during sleep for approximately 4 to 6 weeks to protect the repaired tissues until sufficient healing has occurred.[23,26,36,43,47,48,92,109,174]

### Exercise

The guidelines for progression of exercises during each phase of rehabilitation after shoulder arthroplasty (TSR or hemiarthroplasty) presented in this section are drawn from the limited number of published protocols available, all of which are based on clinical experience rather than evidence from controlled studies and none of which has been shown to be more effective than another.* Also, almost all of these protocols are time-based, with few criteria reported for advancing a patient from one phase of rehabilitation to the next.

Recently, however, several resources have suggested such criteria.[37,47,195] Criteria from these resources are identified for the phases of rehabilitation after shoulder arthroplasty described below. It is important to note that these criteria and suggested timelines for progression of exercises and functional activities are typically adjusted for each patient based on periodic evaluations of the patient's status and ongoing communication between the therapist and the surgeon.

NOTE: The exercise guidelines in this section are for patients *without* preoperative rotator cuff deficiency and who *did not* undergo a cuff repair during TSR or hemiarthroplasty. For a complete picture, modifications in guidelines are noted throughout this section for patients with a poor quality rotator cuff mechanism. Remember, pain relief is the primary goal of shoulder arthroplasty, with improvement in functional mobility a secondary goal. Although improvements in surgical techniques and implant technology now allow more accelerated progression of postoperative rehabilitation than several decades ago, it is still important to proceed judiciously during each phase of rehabilitation to avoid excessive muscle fatigue or irritation or any damage of the healing soft tissues.

### Exercise: Maximum Protection Phase

The maximum protection phase of rehabilitation begins on the first postoperative day and extends for up to 6 weeks. The emphasis of this first phase is patient education, pain control, and initiation of ROM exercises to prevent adhesions and restore shoulder mobility as early as possible to the ranges achieved during surgery. Early motion is permissible after uncemented and cemented shoulder arthroplasty.

While the patient is hospitalized (usually for 3 to 4 days after surgery), patient education includes reviewing early postoperative precautions and teaching the initial exercises in the patient's home program. Precautions during the first 6 weeks after surgery, when protection of soft tissues as they heal is crucial, are summarized in Box 17.6. A patient's adherence to these precautions is of the utmost importance during this phase of rehabilitation.

***Goals and interventions.*** The first phase of rehabilitation includes the following.[23,26,37,43,47,48,91,92,109,174]

**BOX 17.6 Precautions for the Maximum Protection Phase of Rehabilitation After Shoulder Arthroplasty**

**Exercise**

- Short but frequent exercise sessions (four or five times per day).
- Low number of repetitions per exercise.
- Only *passive* or *assisted* shoulder ROM exercises and only within the "safe" limits of ranges noted during surgery. Absolutely no end-range stretching.
  - Passive external rotation to neutral or to less than 30° to avoid excessive stress to the subscapularis muscle, which was incised for surgical exposure and then repaired or lengthened if a preoperative contracture limited external rotation.
  - During passive or assisted shoulder rotation with the patient lying supine, position the humerus slightly anterior to the midline of the body (by placing the arm on a folded towel) to avoid excessive stress to the anterior capsule and suture line.
  - No hyperextension or horizontal abduction (beyond neutral) of the shoulder to avoid stress to the anterior capsule.
  - If an overhead rope-pulley system is used for assisted elevation of the arm, initially *have the patient face the doorway and pulley apparatus* so shoulder elevation occurs only within a limited range.
- Maintain an erect trunk during passive or assisted elevation of the arm while sitting or standing to avoid subacromial impingement of soft tissues.
- In most instances no active (unassisted), antigravity, dynamic shoulder exercises, particularly resisted internal rotation.
- No resistance (strengthening) exercises.
- In general, a more gradual progression of exercises for a patient with a severely damaged and repaired or an irreparable rotator cuff mechanism than a patient with a preoperatively intact cuff.

**Activities of Daily Living**

- Limit activities to those that can be performed with the elbow at waist level, such as eating or writing.
- Avoid reaching behind the back.
- Avoid weight bearing (leaning) on the operated extremity, such as pushing during transfers or when moving in bed, especially the first week after surgery.
- Avoid lifting objects.
- Support the arm in a sling during extended periods of standing or walking.
- Wear the sling while sleeping or outside in crowded areas.
- No driving for several weeks.

*See references 23, 26, 37, 43, 47, 91, 92, 95, 109, 174, 195.

- ***Control pain and inflammation.***
  - Use of a sling or splint for comfort.
  - Use of prescribed analgesic and anti-inflammatory medication.
  - Use of cryotherapy, especially after exercise.
- ***Maintain mobility of adjacent joints.***
  - Active movements of the neck and scapula (while wearing the shoulder immobilizer and after it can be removed for exercise) to maintain normal motion and minimize muscle guarding and spasm. Incorporate "shoulder rolls" by elevating, adducting, and then relaxing the scapulae to reinforce an erect posture of the trunk. Emphasize active scapular retraction and spinal extension.
  - Active ROM of the hand, wrist, and elbow when the arm can be removed from the sling.
- ***Restore shoulder mobility.***
  - Passive or therapist-assisted shoulder motions *within the safe limits determined during surgery.* With the patient lying supine and the arm slightly away from the side of the trunk on a folded towel and the elbow flexed, perform forward elevation of the arm in the plane of the scapula to tolerance, external rotation to no more than 30° to 45°, and internal rotation until the forearm rests on the chest.
  - Pendulum (Codman's) exercises with the elbow flexed (for a shorter moment arm). Encourage the patient to periodically remove the sling and gently swing the arm during ambulation at home.
  - Later during this phase, progress to *self-assisted* shoulder ROM (elevation and rotation) in the *supine position* initially by assisting with the sound hand and later using a wand or dowel rod. Add horizontal abduction to neutral and adduction across the chest holding a wand.
  - Self-assisted shoulder ROM in a *sitting* position or *standing* with a wand by performing "gear shift" exercises (see Fig. 17.23), resting the arm on a table and sliding it forward (see Fig. 17.25), or use of an overhead rope-pulley system to lessen the weight of the arm. Remind the patient to maintain an erect trunk when performing assisted shoulder motions while seated or standing.
  - Self-assisted reaching movements (to the nose, forehead, or over the head as comfort allows) to simulate functional movements.
  - In selected patients, transition to *active* (unassisted) shoulder ROM is often possible by 4 weeks.
  - Functional activities with the elbow at waist level, such as hand to face and writing, are permissible.
- ***Minimize muscle inhibition, guarding, and atrophy.***
  - Gentle muscle-setting of shoulder musculature (excluding the internal rotators) with the elbow flexed and the shoulder in the plane of the scapula or neutral. Teach these exercises prior to discharge from the hospital by having the patient practice isometrically contracting the muscles of the *sound* shoulder. Postpone setting exercises (light isometrics) of the operated shoulder until about 4 to 6 weeks after surgery.
  - Scapular stabilization exercises in non-weight-bearing positions. Target the serratus anterior and trapezius muscles.

NOTE: For a patient who underwent repair of a large tear or rupture of a rotator cuff tendon or other soft tissue reconstruction, it may not be permissible to begin ROM exercises immediately after surgery. When the sling or splint can be removed for exercise, perform only passive or assisted ROM throughout the first phase of rehabilitation. The range of shoulder elevation and external rotation initially permitted may be less than for shoulders that did not require cuff repair. Postpone active (unassisted), antigravity ROM and light isometrics until the next phase (until about 6 weeks postoperatively, when repaired soft tissues are reasonably well healed).

### Exercise: Moderate Protection/ Controlled Motion Phase

Although suggested timelines vary from one resource to another, the moderate protection/controlled motion phase of rehabilitation, which typically begins at about 4 to 6 weeks and extends to at least 12 to 16 weeks or longer postoperatively, focuses on gradually establishing active (unassisted) control, dynamic stability, and strength of the shoulder while continuing to increase ROM.[23,26,43,47,91,92,109,174,195]

***Criteria for progression***. Criteria to advance to the second phase of rehabilitation are:

- ROM: At least 90° of passive elevation, at least 45° degrees of external rotation, and 70° of internal rotation in the plane of the scapula with minimal pain[195] or *almost* full, passive shoulder motion based on intraoperative measurements with little to no pain.[37,92]
- When testing the subscapularis, no tendon pain during resisted, isometric internal rotation.[37]
- Ability to perform most waist-level activities of daily living (ADLs) without pain.[92]

PRECAUTIONS: During this phase of rehabilitation, although it is safe to place increasing stresses (stretching or resistance) on periarticular soft tissues, it is important to do so gradually so as not to irritate these tissues, which are continuing to heal. Therefore, continue with short but frequent exercise sessions (preceded by application of heat and followed by cold) and avoid vigorous stretching or resistance exercises or overuse of the involved shoulder during functional activities.

***Goals and interventions***. The goals and exercises for this next phase of rehabilitation are as follows.

- ***Continue to increase passive ROM of the shoulder.***
  - Transition from passive or assisted ROM to low-*intensity, pain-free* stretching in all anatomical and diagonal planes of motion to achieve full, intraoperative ROM.
  - Gentle joint mobilization techniques for specific capsular restriction.

- To increase shoulder elevation: arm-sliding exercises on the surface of a table and wall-climbing exercises, emphasizing overhead reaching.
- To increase external rotation: With the arm at the side, elbow flexed to 90°, and grasping a doorframe, rotate the trunk away from the involved arm.
- To increase shoulder extension and internal rotation: Add wand exercises behind the back. (This exercise also involves scapular winging and tipping, both necessary for reaching behind the back.)
- To increase horizontal adduction: Cross-body stretch with arm at shoulder level.

- ***Develop active control and dynamic stability and improve muscle performance (strength and endurance) of the shoulder.***
  - Continue or gradually transition to *active* shoulder ROM exercises, avoiding antigravity abduction until the patient can initiate the movement without first shrugging the operated shoulder.
  - Scapular and GH joint stabilization exercises (alternating isometrics and rhythmic stabilization) in a variety of positions initially in non-weight-bearing positions, progresing to light weight bearing.
  - Pain-free, low-intensity (submaximal) resisted isometrics of shoulder muscles, including the subscapularis or any other repaired muscle-tendon units.
  - Dynamic resistance exercises for the scapula and shoulder musculature (from 0° to 90° of shoulder elevation) using light weights or light-grade elastic resistance. Begin in the supine position to support and stabilize the scapula. Progress to the sitting position. Place emphasis on improving control of the rotator cuff to prevent superior translation of the humerus during shoulder elevation.[166]
  - Upper extremity ergometry with stationary ergometer or a portable reciprocal exerciser on a table. Emphasize progressive repetitions to increase muscular endurance.

### Exercise: Minimum Protection/ Return to Functional Activity Phase

The minimum protection/return to functional activity phase usually begins around 12 to 16 weeks postoperatively (depending on rotator cuff tissue quality and function) and typically extends for several more months.[47,92,195] There is a continued effort during this phase to restore full, active ROM for functional activities so long as there is adequate stability of the GH joint. Consequently, pain-free strengthening of the shoulder girdle for dynamic stability and functional use of the upper extremity for progressively more demanding tasks is the primary focus of this phase. For optimal results, the home exercise program may need to be continued for 6 months or longer, and functional and recreational activities may need to be modified.

***Criteria for progression***. To advance to the final phase of rehabilitation and gradually return to necessary and desired functional activities, a patient should meet the following criteria.

- Full, passive ROM of the shoulder (based on intraoperative ranges)[37,92] or at least 130° to 140° of pain-free, passive or assisted shoulder flexion and 120° of abduction.[195]
- In the plane of the scapula, at least 60° pain-free, passive external rotation and 70° internal rotation.[195]
- Active (unassisted), antigravity elevation of the arm to at least 100° to 120° in the plane of the scapula while maintaining joint stability and using appropriate shoulder mechanics, particularly no shoulder shrugging prior to elevating the arm.[195]
- Strength of rotator cuff and deltoid muscles: 4/5.[47,92]

***Goals and interventions***. Goals and activities for the final phase of rehabilitation include the following.[23,26,37,47,48,91,92,195]

- ***Continue to improve or maintain shoulder mobility.***
  - End-range self-stretching.
  - Grade III joint mobilization and self-mobilization, if appropriate.[23,26,37,92]
- ***Continue to improve active control and muscle performance of the shoulder.***
  - Pain-free, low-load, high-repetition progressive resistive exercise (PRE) of shoulder musculature in anatomical and diagonal planes and in patterns of movement that replicate functional tasks throughout the available ROM. Position the patient in a variety of gravity-resisted positions.
  - Closed-chain, resisted shoulder exercises, gradually increasing weight bearing through the upper extremity.
  - Use of the involved upper extremity for lifting, carrying, pushing, or pulling activities against increasing loads.
- ***Return to most functional activities.***
  - Use of the operated upper extremity for progressively more advanced functional activities.
  - Recreational activities, such as swimming and golf are possible.
  - Modification of high-demand, high-impact work-related or recreational activities to avoid imposing excessive forces on the GH joint that could lead to loosening or premature wear of prosthetic implants.

NOTE: For the patient whose rotator cuff was irreparable or continues to be significantly deficient because of a tenuous repair and who has limited but pain-free shoulder ROM, modification of the environment and use of assistive devices may be necessary for independence in functional activities.

### Outcomes

Over the past 30 years, as patient selection criteria, prosthetic designs, and surgical techniques have been refined, postoperative outcomes after shoulder arthroplasty have improved. As noted at the beginning of this section on glenohumeral arthroplasty, numerous resources suggest that outcomes after TSR or hemiarthroplasty are influenced by many factors, including the type and severity of the underlying pathology, the status of soft tissues (espe-

cially the rotator cuff mechanism and subscapularis), the type and quality of the surgical procedure(s) performed, and patient-related factors, such as participation in a postoperative rehabilitation program.[32,36,195] The outcomes most often measured in follow-up studies using a variety of quantitative scales are pain relief and other paramenters of quality of life, passive and active shoulder ROM, and the ability to perform functional activities.

Despite the emphasis in numerous resources that a patient's participation in postoperative rehabilitation is crucial for successful outcomes, there are no studies to support this opinion because all patients undergoing shoulder arthroplasty are given some form of postoperative exercise instruction. Furthermore, published protocols are routinely modified to meet the needs of individual patients and consequently have not been compared to determine if one protocol yields better outomes than another.[195]

***Pain relief.*** A decrease in pain is the most dramatic result of glenohumeral arthroplasty. Almost all patients—regardless of the underlying pathology, the type of arthroplasty (TSR or hemireplacement), or the design of the prosthetic implants—report complete or substantial relief of shoulder pain and improved functional use of the arm.[36,108,109,117,119,121,130,134,171]

The extent of pain relief has been shown to be associated with the underlying cause(s) of glenohumeral arthritis. Neer et al.,[119] Matsen,[108] and more recently Norris and Iannotti[130] reported that 90% of patients with primary OA or osteonecrosis had complete or near-complete pain relief after TSR. Similar results have been reported for patients with OA who underwent hemiarthroplasty.[101,109,121] Patients with RA or other synovium-based diseases also report substantial pain relief after TSR or hemiarthroplasty, although not quite to the extent reported by patients with OA or osteonecrosis.[36,159,182]

Whether TSR is more effective than hemiarthroplasty for pain relief has also been studied. In a prospective follow-up study over a mean of 4.3 years when patients with OA having TSR were compared to those having hemiarthroplasty, postoperative pain scores were reported to be similar in the two groups, with patients in the TSR group demonstrating more improvement because of a higher level of pain preoperatively.[134] In another study, patients with OA were randomly assigned to undergo either TSR or hemiarthroplasty and were evaluated postoperatively over a 24-month period. Results of this study indicated that both groups of patients reported significant pain relief and improvements in other quality-of-life parameters, with no significant differences between the TSR and hemiarthroplasty groups.[101] Whether TSR versus hemiarthroplasty is more effective for pain relief in patients with RA has not been clearly established.[109,182]

***ROM and functional use of the upper extremity.*** Despite the emphasis placed on improving ROM and use of the arm for function activities during rehabilitation after shoulder arthroplasty, improvements in these outcomes are less predictable than pain relief, with the functional status improving more consistently than ROM.* In general, patients with primary OA or osteonecrosis demonstrate greater improvement in active ROM (forward elevation and shoulder rotation) than patients with RA, in part because of a higher incidence of cuff deficiency asociated with RA or the use of more constrained prosthetic designs.[159,182,195] For example, in patients with OA or osteonecrosis, the mean active forward elevation of the shoulder (reported in reviews of a number of studies) changed from 105° to 161°. In patients with RA means ranged from 75° to 105°.[171,195]

Significant improvement in functional status has been reported for patients with OA or osteonecrosis. Although functional improvement after arthroplasty has been reported for patients with RA, many studies used nonstandardized measurement tools, making it difficult to compare their results with those of other studies.[195] Regardless of the underlying pathology, resources agree that a well functioning rotator cuff mechanism is the basis for significant postoperative gains in active ROM and functional abilities.[36,168,195]

## Arthrodesis of the Shoulder

### Indications for Surgery

The following are generally accepted indications for arthrodesis of the GH joint.[109,145,146,196]

- Incapacitating pain (at rest or with movement)
- Gross instability of the GH joint
- Complete paralysis of the deltoid and rotator cuff muscles
- Severe joint destruction due to infection
- Failed shoulder arthroplasty in a young, active patient who is not a candidate for revision arthroplasty
- Sufficient compensatory scapular motion and strength of the serratus anterior and trapezius muscles

### Procedure

The GH joint is fused with pins and bone grafts in a position of 15° to 30° flexion and abduction and up to 30° to 45° internal rotation so the hand can reach the middle of the body or the mouth.[109,145,146,174,196] The shoulder is immobilized in a shoulder spica cast or a thoracobrachial (airplane) splint that extends across the elbow joint. The immobilizer is worn for 3 to 5 months.

### Postoperative Management

Place emphasis on maintaining mobility of peripheral joints (wrist and hand) while the shoulder and elbow are immobilized. If a splint with a hinged elbow joint is used, begin elbow flexion and extension when permis-

*See references 57, 109, 130, 134, 161, 171, 195.